LAKES / INDIA LAKES AND WETLANDS

CATEGORY BOUNDARY → LAKE-ORIGIN TREE → WATER-BUDGET EQUATION → LAKE-CHANGE FIREWALL → WETLAND HYDROPERIOD → INDIA LAKE MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CATEGORY BOUNDARY

CATEGORY BOUNDARY | STANDING WATER IN AN INLAND BASIN | ARTIFICIAL OR STRONGLY REGULATED STORAGE | BROADER HYDROLOGICAL-ECOLOGICAL CATEGORY | RAMSAR SITE | INTERNATIONAL DESIGNATION, NOT BASIN ORIGIN | MUST REMEMBER | LAKE CLASSIFICATION MUST FOLLOW BASIN ORIGIN—TECTONIC, VOLCANIC, GLACIAL, FLUVIAL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
LAKE → standing water in an inland basin	WETLAND → broader hydrological-ecological category	MUST REMEMBER: Lake classification must follow basin origin—tectonic, volcanic, glacial, fluvial, karst, coastal, aeolian, landslide or artificial—then water
RESERVOIR → artificial or strongly regulated storage	RAMSAR SITE → international designation, not basin origin	Indian wetlands add Ramsar and domestic-governance layers.

CORE CLAIM
- LAKE → standing water in an inland basin
- RESERVOIR → artificial or strongly regulated storage

MECHANISM
- WETLAND → broader hydrological-ecological category
- RAMSAR SITE → international designation, not basin origin

CONSEQUENCE / LIMIT
- MUST REMEMBER: Lake classification must follow basin origin—tectonic, volcanic, glacial, fluvial, karst, coastal, aeolian, landslide or artificial—then water
- Indian wetlands add Ramsar and domestic-governance layers.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Lake classification must follow basin origin—tectonic, volcanic, glacial, fluvial, karst, coastal, aeolian, landslide or artificial—then water balance, mixing, sediment and succession; Indian wetlands add Ramsar and domestic-governance layers.

01

STAGE 01 LAKE-ORIGIN TREE

LAKE-ORIGIN TREE | EARTH MOVEMENT | TECTONIC OR VOLCANIC BASIN | TARN, OX-BOW OR LAGOON | LANDSLIDE DAM OR CRATER | HUMAN WORK | RESERVOIR, BUND OR EXCAVATION

EARTH MOVEMENT → tectonic or volcanic basin

ICE / RIVER / COAST → tarn, ox-bow or lagoon → BLOCKAGE / IMPACT → landslide dam or crater → HUMAN WORK → reservoir, bund or excavation

CORE CLAIM
- EARTH MOVEMENT → tectonic or volcanic basin
- ICE / RIVER / COAST → tarn, ox-bow or lagoon

MECHANISM
- BLOCKAGE / IMPACT → landslide dam or crater
- HUMAN WORK → reservoir, bund or excavation

CONSEQUENCE / LIMIT
- EARTH MOVEMENT → tectonic or volcanic basin
- ICE / RIVER / COAST → tarn, ox-bow or lagoon

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLOCKAGE / IMPACT → landslide dam or crater.

02

STAGE 02 WATER-BUDGET EQUATION

WATER-BUDGET EQUATION | RAIN + RIVER + GROUNDWATER | EVAPORATION + OUTFLOW + SEEPAGE | STORAGE CHANGE | INPUTS MINUS OUTPUTS | STATE PERIOD, BASIN AND UNCERTAINTY

INPUT → rain + river + groundwater → OUTPUT → evaporation + outflow + seepage → STORAGE CHANGE → inputs minus outputs → ATTRIBUTION → state period, basin and uncertainty

CORE CLAIM
- INPUT → rain + river + groundwater
- OUTPUT → evaporation + outflow + seepage

MECHANISM
- STORAGE CHANGE → inputs minus outputs
- ATTRIBUTION → state period, basin and uncertainty

CONSEQUENCE / LIMIT
- INPUT → rain + river + groundwater
- OUTPUT → evaporation + outflow + seepage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE CHANGE → inputs minus outputs; ATTRIBUTION → state period, basin and uncertainty.

03

STAGE 03 LAKE-CHANGE FIREWALL

LAKE-CHANGE FIREWALL | SEDIMENT AND ORGANIC INFILLING | NUTRIENT-PRODUCTIVITY-OXYGEN SHIFT | WATER-BUDGET DEFICIT | SIMILAR APPEARANCE CAN HIDE DIFFERENT MECHANISMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT

CORE CLAIM
- INPUT → rain + river + groundwater
- OUTPUT → evaporation + outflow + seepage

MECHANISM
- STORAGE CHANGE → inputs minus outputs
- ATTRIBUTION → state period, basin and uncertainty

CONSEQUENCE / LIMIT
- INPUT → rain + river + groundwater
- OUTPUT → evaporation + outflow + seepage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE CHANGE → inputs minus outputs; ATTRIBUTION → state period, basin and uncertainty.

03

STAGE 03 LAKE-CHANGE FIREWALL

LAKE-CHANGE FIREWALL | SEDIMENT AND ORGANIC INFILLING | NUTRIENT-PRODUCTIVITY-OXYGEN SHIFT | WATER-BUDGET DEFICIT | SIMILAR APPEARANCE CAN HIDE DIFFERENT MECHANISMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SUCCESSION → sediment and organic infilling	SHRINKAGE → water-budget deficit	SUCCESSION → sediment and organic infilling
EUTROPHICATION → nutrient-productivity-oxygen shift	RULE → similar appearance can hide different mechanisms	EUTROPHICATION → nutrient-productivity-oxygen shift

CORE CLAIM
- SUCCESSION → sediment and organic infilling
- EUTROPHICATION → nutrient-productivity-oxygen shift

MECHANISM
- SHRINKAGE → water-budget deficit
- RULE → similar appearance can hide different mechanisms

CONSEQUENCE / LIMIT
- SUCCESSION → sediment and organic infilling
- EUTROPHICATION → nutrient-productivity-oxygen shift

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → similar appearance can hide different mechanisms.

04

STAGE 04 WETLAND HYDROPERIOD

WETLAND HYDROPERIOD | SEASON OF INUNDATION | HOW LONG WATER REMAINS | DEPTH + FREQUENCY | HABITAT FILTER | RIVER, AQUIFER OR COAST LINK

TIMING → season of inundation → DURATION → how long water remains → DEPTH + FREQUENCY → habitat filter → CONNECTIVITY → river, aquifer or coast link

CORE CLAIM
- TIMING → season of inundation
- DURATION → how long water remains

MECHANISM
- DEPTH + FREQUENCY → habitat filter
- CONNECTIVITY → river, aquifer or coast link

CONSEQUENCE / LIMIT
- TIMING → season of inundation
- DURATION → how long water remains

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEPTH + FREQUENCY → habitat filter; CONNECTIVITY → river, aquifer or coast link.

05

STAGE 05 INDIA LAKE MAP

INDIA LAKE MAP | JHELUM-LINKED KASHMIR BASIN | SALINE RAJASTHAN CLOSED BASIN | ODISHA COASTAL LAGOON

WULAR → Jhelum-linked Kashmir basin

SAMBHAR → saline Rajasthan closed basin → CHILIKA → Odisha coastal lagoon → LONAR / KANWAR / BHOJ → impact / fluvial / artificial

CORE CLAIM
- WULAR → Jhelum-linked Kashmir basin
- SAMBHAR → saline Rajasthan closed basin

MECHANISM
- CHILIKA → Odisha coastal lagoon
- LONAR / KANWAR / BHOJ → impact / fluvial / artificial

CONSEQUENCE / LIMIT
- WULAR → Jhelum-linked Kashmir basin
- SAMBHAR → saline Rajasthan closed basin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONAR / KANWAR / BHOJ → impact / fluvial / artificial.

06

STAGE 06 RIVER-LAKE TRAPS

RIVER-LAKE TRAPS | LINKED TO JHELUM | BETWEEN KRISHNA AND GODAVARI DELTAS | PHUMDI-BEARING LAKE | ADJACENCY DOES NOT PROVE DIRECT FEEDING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
WULAR → linked to Jhelum	LOKTAK → phumdi-bearing lake	WULAR → linked to Jhelum
KOLLERU → between Krishna and Godavari deltas	RULE → adjacency does not prove direct feeding	KOLLERU → between Krishna and Godavari deltas

CORE CLAIM
- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

MECHANISM
- LOKTAK → phumdi-bearing lake
- RULE → adjacency does not prove direct feeding

CONSEQUENCE / LIMIT
- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

06 — STAGE 06 RIVER-LAKE TRAPS

RIVER-LAKE TRAPS · LINKED TO JHELUM · BETWEEN KRISHNA AND GODAVARI DELTAS · PHUMDI-BEARING LAKE · ADJACENCY DOES NOT PROVE DIRECT FEEDING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
WULAR → linked to Jhelum	LOKTAK → phumdi-bearing lake	WULAR → linked to Jhelum
KOLLERU → between Krishna and Godavari deltas	RULE → adjacency does not prove direct feeding	KOLLERU → between Krishna and Godavari deltas

CORE CLAIM
- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

MECHANISM
- LOKTAK → phumdi-bearing lake
- RULE → adjacency does not prove direct feeding

CONSEQUENCE / LIMIT
- WULAR → linked to Jhelum
- KOLLERU → between Krishna and Godavari deltas

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjacency does not prove direct feeding.

07 — STAGE 07 URBAN RECLAMATION CHAIN

URBAN RECLAMATION CHAIN · STORAGE AND DRAINAGE LOST · FLOOD PEAK AND DURATION RISE · DILUTION AND HABITAT DECLINE · LOW-LYING COMMUNITIES BEAR SHIFTED RISK

FILL / NARROW → storage and drainage lost → RUNOFF → flood peak and duration rise → POLLUTION → dilution and habitat decline → EQUITY → low-lying communities bear shifted risk

CORE CLAIM
- FILL / NARROW → storage and drainage lost
- RUNOFF → flood peak and duration rise

MECHANISM
- POLLUTION → dilution and habitat decline
- EQUITY → low-lying communities bear shifted risk

CONSEQUENCE / LIMIT
- FILL / NARROW → storage and drainage lost
- RUNOFF → flood peak and duration rise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FILL / NARROW → storage and drainage lost.

08 — STAGE 08 THREAT-RESPONSE MATRIX

THREAT-RESPONSE MATRIX · SOURCE CONTROL + TREATMENT · CATCHMENT AND EROSION MANAGEMENT · PROTECT HYDROLOGICAL SPACE · ALTERED FLOW · ECOLOGICAL AND FLOW RESTORATION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
SEWAGE / NUTRIENT → source control + treatment	ENCROACHMENT → protect hydrological space	SEWAGE / NUTRIENT → source control + treatment
SEDIMENT → catchment and erosion management	INVASIVE / ALTERED FLOW → ecological and flow restoration	SEDIMENT → catchment and erosion management

CORE CLAIM
- SEWAGE / NUTRIENT → source control + treatment
- SEDIMENT → catchment and erosion management

MECHANISM
- ENCROACHMENT → protect hydrological space
- INVASIVE / ALTERED FLOW → ecological and flow restoration

CONSEQUENCE / LIMIT
- SEWAGE / NUTRIENT → source control + treatment
- SEDIMENT → catchment and erosion management

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INVASIVE / ALTERED FLOW → ecological and flow restoration.

09 — STAGE 09 RAMSAR STATUS FIREWALL

RAMSAR STATUS FIREWALL · AT LEAST ONE CAN SUPPORT LISTING · WISE USE · MAINTAIN ECOLOGICAL CHARACTER · PRIORITY ATTENTION TO CHARACTER CHANGE · LISTING IS NOT AUTOMATIC NATIONAL PARK STATUS · CLOSE DISTINCTION · LAKE, LAGOON, OXBOW, RESERVOIR AND WETLAND ARE NOT SYNONYMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CRITERION → at least one can support listing	LIMIT → listing is not automatic National Park status	international recognition rather than ownership transfer or automatic protection.
WISE USE → maintain ecological character	CLOSE DISTINCTION: Lake, lagoon, oxbow, reservoir and wetland are not synonyms	
MONTREUX → priority attention to character change	freshwater/saline and exorheic/endorheic are separate axes, while Ramsar designation is	

CORE CLAIM
- CRITERION → at least one can support listing
- WISE USE → maintain ecological character
- MONTREUX → priority attention to character change

MECHANISM
- LIMIT → listing is not automatic National Park status
- CLOSE DISTINCTION: Lake, lagoon, oxbow, reservoir and wetland are not synonyms
- freshwater/saline and exorheic/endorheic are separate axes, while Ramsar designation is

CONSEQUENCE / LIMIT
- international recognition rather than ownership transfer or automatic protection.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: freshwater/saline and exorheic/endorheic are separate axes, while Ramsar designation is international recognition rather than ownership transfer or automatic protection.

RAMSAR STATUS FIREWALL | AT LEAST ONE CAN SUPPORT LISTING | WISE USE | MAINTAIN ECOLOGICAL CHARACTER | PRIORITY ATTENTION TO CHARACTER CHANGE | LISTING IS NOT AUTOMATIC NATIONAL PARK STATUS | CLOSE DISTINCTION

LAKE, LAGOON, OXBOW, RESERVOIR AND WETLAND ARE NOT SYNONYMS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CRITERION → at least one can support listing	LIMIT → listing is not automatic National Park status	international recognition rather than ownership transfer or automatic protection.
WISE USE → maintain ecological character	CLOSE DISTINCTION: Lake, lagoon, oxbow, reservoir and wetland are not synonyms	
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CORE CLAIM

- CRITERION → at least one can support listing
- WISE USE → maintain ecological character
- MONTREUX → priority attention to character change

MECHANISM

- LIMIT → listing is not automatic National Park status
- CLOSE DISTINCTION: Lake, lagoon, oxbow, reservoir and wetland are not synonyms
- freshwater/saline and exorheic/endorheic are separate axes, while Ramsar designation is

CONSEQUENCE / LIMIT

- international recognition rather than ownership transfer or automatic protection.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: freshwater/saline and exorheic/endorheic are separate axes, while Ramsar designation is international recognition rather than ownership transfer or automatic protection.

10

STAGE 10 INDIA GOVERNANCE CHAIN

INDIA GOVERNANCE CHAIN | IDENTIFY AND DELINEATE SYSTEM | REGULATE ACTIVITIES | FINANCE AND IMPLEMENT MANAGEMENT | PROVE ECOLOGICAL RESPONSE | EVIDENCE LIMIT | AREA, DESIGNATION COUNT AND ECOLOGICAL STATUS ARE TIME-SENSITIVE; SHRINKAGE

INVENTORY → identify and delineate system

NOTIFICATION / RULES → regulate activities → PLAN / NPCA → finance and implement management → MONITOR → prove ecological response

CORE CLAIM

- INVENTORY → identify and delineate system
- NOTIFICATION / RULES → regulate activities

MECHANISM

- PLAN / NPCA → finance and implement management
- MONITOR → prove ecological response

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Area, designation count and ecological status are time-sensitive
- shrinkage or eutrophication needs inflow, evaporation, abstraction, nutrients, encroachment and catchment evidence rather than one-cause attribution.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Area, designation count and ecological status are time-sensitive; shrinkage or eutrophication needs inflow, evaporation, abstraction, nutrients, encroachment and catchment evidence rather than one-cause attribution.

11

STAGE 11 LAKE-WETLAND ANSWER SPINE

LAKE-WETLAND ANSWER SPINE | BASIN, WETLAND AND STATUS | ORIGIN AND HYDROLOGY | BUDGET, NUTRIENT OR CONNECTIVITY FAILURE | BASIN-SCALE, MONITORED AND STATUS-AWARE

DEFINE → basin, wetland and status → CLASSIFY → origin and hydrology → DIAGNOSE → budget, nutrient or connectivity failure → RESPOND → basin-scale, monitored and status-aware

CORE CLAIM

- DEFINE → basin, wetland and status
- CLASSIFY → origin and hydrology

MECHANISM

- DIAGNOSE → budget, nutrient or connectivity failure
- RESPOND → basin-scale, monitored and status-aware

CONSEQUENCE / LIMIT

- DEFINE → basin, wetland and status
- CLASSIFY → origin and hydrology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIAGNOSE → budget, nutrient or connectivity failure; RESPOND → basin-scale, monitored and status-aware.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | INDIAN GLACIAL LAKES IN THE SOURCE ARE HIMALAYAN | GANGABAL, SHESHNAG, KAILASH-KUND, SANASAR. | NO PLEISTOCENE GLACIATION IN MAJID'S ACCOUNT; HENCE NO PENINSULAR | WULAR (J&K) | TECTONIC FRESHWATER LAKE; AMONG ASIA'S LARGEST FRESHWATER LAKES. | CHILIKA AND PULICAT | EAST-COAST LAGOONS.

OPTIONAL SOURCE DEPTH

- Indian glacial lakes in the source are Himalayan: Gangabal, Sheshnag, Kailash-Kund, Sanasar.
- Peninsula: no Pleistocene glaciation in Majid's account; hence no Peninsular glacial tarn belt.
- Wular (J&K) = tectonic freshwater lake; among Asia's largest freshwater lakes.
- Chilika and Pulicat = east-coast lagoons.

ADVANCED DEBATE

- Sambhar (Rajasthan) = major inland salt lake.
- Sasthamkotta = Kerala's largest freshwater lake.
- Sat Tal = Kumaun lake group near Bhimtal; avoid reducing its origin to a single glacial label.
- NOT: Peninsular India has glacial tarn lakes → Glacial lakes are Himalayan in the source; Peninsular India shows no Pleistocene glaciation .

LEGACY / NUANCE

- NOT: Sambhar is a freshwater lake → Sambhar is a salt lake of Rajasthan's arid inland drainage.
- NOT: Pulicat is a west-coast lagoon → Pulicat is on the east coast (AP/TN).
- NOT: Wular is a lagoon → Wular is tectonic/freshwater , not coastal lagoonal.
- NOT: All Ramsar lakes have the same origin → Ramsar status is conservation status; origin may be glacial, tectonic, lagoonal, man-made or salt.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels